Bayesian data analysis mini project

This exercise uses a sample dataset from the *agridat* package to demonstrate the process of building and interpreting Bayesian linear models in R. The dataset is from a real study of sheep breeding. Page 2 has a causal diagram illustrating the variables in the dataset. You can learn more about the dataset by loading the agridat library and running the command ?ilri.sheep in the console. You can also view the documentation here: https://www.rdocumentation.org/packages/agridat/versions/1.23/topics/ilri.sheep.

The exercises are contained in two scripts: sheepModelPart1.R and sheepModelPart2.R. To do the exercises, download the scripts from the github repository (https://github.com/DRIVES-Project/CANVAS2026_BayesWorkshop/tree/main/miniProject) and run through the code and follow the prompts in the comments. The first part goes through the process of doing exploratory analysis to develop hypotheses and using frequentist tools to build and compare linear models. The second part builds models with Bayesian tools (i.e., *brms*) and compares them to frequentist results.

Because the code is provided, the purpose of this exercise is to, 1) run through the steps of a Bayesian data analysis and interpret the results, and 2) familiarize yourself with coding tools that you can repurpose for your own analysis. Note: this exercise does not include fitting and comparing multiple Bayesian models. For a deep dive on this topic, check out https://arviz-devs.github.io/EABM/Chapters/Model_comparison.html.

Use the prompts and sample code to:

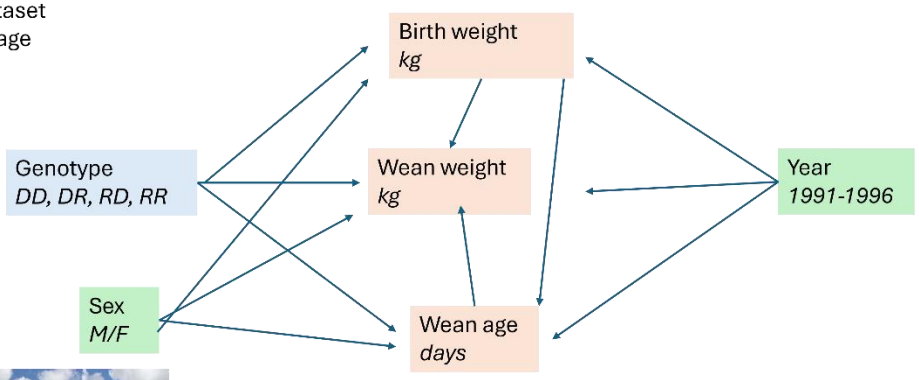
1) Do exploratory analysis (i.e., data visualization and summary stats) to help prepare for statistical analysis.
2) Build frequentist linear models to familiarize yourself with model structure in R and help select models to fit with Bayesian methods.
3) Use frequentist model evaluation and interpretation tools that have Bayesian analogs in *brms*.
4) Build and interpret a Bayesian model using brms and compare to the results from frequentist models.

Optional exercise: Write a short results paragraph presenting the results for the Bayesian model.

Birth weight and weaning weight of Dorper x Red Maasi lambs

ilri.sheep dataset
agridat package

How does genotype influence birth weight, weaning weight, and weaning age?





Birth weight
Wean weight
Wean age

F(Genotype, Sex, Year)
F(Genotype, Sex, Year, Birth weight, Wean age)
F(Genotype, Sex, Year, Birth weight)

Treatment

Outcome

Covariate